

Nathaniel H. Brill

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years building enterprise full-stack applications at UnitedHealth Group using React, TypeScript, Java Spring Boot, and Express.js. Delivered scalable healthcare solutions serving thousands of users across multi-state programs while maintaining 99%+ uptime and zero security incidents. Skilled in API optimization, cloud infrastructure (AWS/Azure/GCP), and CI/CD automation with proven ability to solve complex technical challenges.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL, C, Bash

Front End: React, Node.js, Angular, HTML/CSS

Back End: Spring Boot, Express.js, Apache Maven, RESTful APIs, JUnit, LangChain, LangGraph, NumPy, Pandas

Cloud & DevOps: AWS (Certified), GCP, Azure, GitHub Actions, Jenkins, Grafana, Docker, Airflow, Kubernetes

Tools: Git, Microsoft Office, Grafana, Splunk, Rally, Postman, DBeaver, Artifactory, Figma, Unix, Copilot, SonarQube, Agile/Scrum

Certifications: AWS Certified Cloud Practitioner.

EXPERIENCE

UnitedHealth Group, Optum

Phoenix, Arizona

Software Engineer

May 2022 - Present

- Built React frontend and Express.js microservices for Medicare/Medicaid claim correction platform processing ~\$4 billion annually across 12 state programs, enabling users to search, validate, and submit healthcare claims.
- Developed real-time operational dashboards using an internal library built off Chart.js visualizing claim status metrics (accepted, closed, errored), enabling stakeholders to monitor claim processing and generate reports.
- Decreased API response times from over 5s to under 2s optimizing SQL queries and implementing database indexing strategies.
- Remediated 100+ Critical and High CVEs identified with CodeQL, Dependabot, Prism, Maven and SonarQube.
- Achieved over 90% code coverage across 56 repos through comprehensive JUnit test suites.
- Implemented automated claim validation and pre-submission correction workflows using Drools business rules engine, reducing manual review time and preventing submission errors to state systems.
- Architected and deployed CI/CD pipeline using GitHub Actions for Azure Function Apps to replace deprecated Jenkins.
- Documented migration procedures and trained engineering team on Google Anthos platform administration, enabling seamless transition from OpenShift to GCP.
- Set up data streaming for chatbots by coding an event hub and workspace log analytics as IAC in terraform.
- Created a POST API in bash to generate a Grafana dashboard to measure errors.
- Completed advanced technical training in generative AI by building a proof-of-concept claim approval agent tool using Python, LangChain, and LangGraph to explore LLM applications in healthcare automation.

UnitedHealth Group, Optum

Phoenix, Arizona

Software Engineer Intern

Summer 2020 & Summer 2021

- Created a web application timeline of patients' medical history and appointments using Angular and Java Springboot
- Delivered \$526,692 in annual cost savings by architecting an automated task scheduler with Apache Airflow, Python, and Docker Swarm, eliminating inefficient manual legacy system operations.
- Collaborated in an agile team of 5 engineers, presenting milestones to senior level stakeholders and incorporating feedback into sprint planning.

Denver Public Schools, Department of Technology and Services

Denver, Colorado

Software Engineer Intern

Summer 2017

- Built data collection web form with C# controllers and MongoDB backend.
- Implemented responsive UI using Bootstrap and Knockout.js for categorizing fixed assets.

ACTIVITIES

Volunteering: Habitat for Humanity, Midwest Food bank, Feed My Starving Children, Natural Restorations

Organizations: Member of Theta Tau Chi Professional Engineering Fraternity, member of Tau Kappa Epsilon

EDUCATION

University of Arizona

Tucson, Arizona

Bachelor of Science in Computer Science, Minor in Business Administration

Diploma received May 2022